SECTIONS ANNIHILATED BY A FINITE TYPE IDEAL FORM A QUASI-COHERENT SHEAF

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Editorial note. Counted span is the author's own proof environment, source lines 3455-3461, following the statement at 3433-3453.

Editorial note. The 'It is clear' sentence is a definitional subsheaf reading, reproduced in the notes.

Allowed standard prerequisites (editorial).

- Scheme-level sections annihilated by an ideal. Properties, Lemma properties-lemma-sections-annihilated-by-ideal (cross-chapter, cited at ~3459-3461). Concluding engine after etale-local reduction.

Source reference. Original reference 01PO: focused-seeds/geometry/S9-spaces-limits-sections-annihilated-by-ideal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Consider the sheaf of $\mathcal{O}_X$-modules $\mathcal{F}'$ which associates to every object U of $X_{étale}$ the module*

$$\mathcal{F}'(U) = \{s \in \mathcal{F}(U) \mid \mathcal{I}s = 0\}$$

Assume $\mathcal{I}$ is of finite type. Then

1. *$\mathcal{F}'$ is a quasi-coherent sheaf of $\mathcal{O}_X$-modules,*
2. *for affine U in $X_{étale}$ we have $\mathcal{F}'(U) = \{s \in \mathcal{F}(U) \mid \mathcal{I}(U)s = 0\}$, and*
3. *$\mathcal{F}'_x = \{s \in \mathcal{F}_x \mid \mathcal{I}_x s = 0\}$.*

Proof. It is clear that the rule defining $\mathcal{F}'$ gives a subsheaf of $\mathcal{F}$. Hence we may work étale locally on X to verify the other statements. Thus the lemma reduces to the case of schemes which is Properties, Lemma 01PO. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.